

THE PARADOX OF RURAL INNOVATION: THOSE WHO NEED TECHNOLOGY MOST, ACCESS IT LEAST

As a small island nation highly dependent on agriculture and extremely vulnerable to the consequences of climate change, the last decade or more has been a series of relentless shocks to the vulnerable farming communities of Sri Lanka's dry zone. Erratic rainfall, prolonged droughts, higher temperatures, floods, a global pandemic and geo-political crises which though seem far removed from the beautiful shores of Sri Lanka have taken a high toll on the country's agricultural sector. The steep increases in import costs and agricultural inputs have left many farmers struggling to recover from overlapping and interlinked pressures.

For farmers in the North, the strain is often sharper. Energy poverty is higher, and essential technologies that could help buffer climate and economic shocks remain out of reach for many rural producers.



"We are a large-scale farming family," he says.

"Due to this, we had to pay a high amount for electricity. In some months, it became very difficult, and we had to borrow money to pay the electricity bills."

In Kirishnapuram, Kilinochchi, Suntharalingam Jokeswaran is a rare profile in the farming community: an entrepreneurial, tech-savvy graduate from the University of Jaffna who chose to farm his ancestral land instead of leaving his family and home behind in search of more lucrative opportunities in Colombo or abroad. He cultivates passion fruit, papaya, and vegetables, and he speaks about energy the way many speak about rainfall: as something that can make or break an entire season.



Across rural Sri Lanka access to technology and skills training is often limited. For many households, modernization is not a matter of ambition; it is a matter of affordability and opportunity.

Through the 'Empowering Women Smallholders Farmers through Renewable Energy-Based Agriculture Modernization Technologies' project, the United Nations Development Programme (UNDP) in Sri Lanka with funding from the Japanese Supplementary Budget (JSB) of the Government of Japan, beneficiaries like Jokeswaran received a new boost.

He was supported with a solar-powered irrigation system along with the relevant knowledge and capacity, that sprinkled some new hope.

"The main advantage is that it has significantly reduced the electricity cost for our cultivation"

That single change, shifting irrigation to a decentralized renewable source did more than cut bills. It turned energy from a recurring crisis into a reliable input, allowing him to expand cultivation and diversify crops.

"Through this support, we have been able to expand our cultivated land area and grow a variety of crops,"

In a country where crisis periods have repeatedly shown how energy insecurity can quickly impact food and livelihoods, decentralized clean energy offers something rare: reliability.

“Even during fuel crises or electricity shortages, we can continue cultivation without interruption and can increase our production,”

For rural producers, the ability to continue farming through disruption is not only an economic advantage— but it is a safeguard for human security. This is precisely the project's ambition: strengthening energy security to protect livelihoods and human security for climate-vulnerable communities, while laying foundations for a just transition and net-zero pathways in the agriculture sector.

Jokeshwaran's enthusiasm is unmistakable, not only for what the technology does, but for what it represents.

“I am a graduate and an entrepreneur who is interested in exploring new technologies,” he says. “I am proud to be one of the farmers using this renewable energy technology that provides us eco-friendly solutions.”



Jokeshwaran's story stands out from the multitude of others that represent rural farmers as simply the recipients of assistance: it is about how access to the right technology can elevate capable rural innovators into commercially viable producers.

Across Kilinochchi, this project has delivered 100 solar irrigation systems, coupled with maintenance and management training, SHEP (Smallholder Horticulture Empowerment and Promotion) -based crop planning and marketing training, and financial literacy training — a package designed to ensure that clean energy systems translate into lasting development outcomes, not short-term relief.





In Kirishnapuram, the proof is already visible: technology has become the development enabler, and in times of instability, it can be the difference between halting cultivation and sustaining it.